TRACK 1: AGENTIC CYBER DEFENCE • HACK@AI 2026

Design and Development of a Post-Quantum Cryptography-Based Secure Code Review Tool

From Alert Overload to Explainable, Controlled Post-Quantum Remediation



🛡️ 01. Take In Evidence

Deterministic AST code ingestion replaces noisy regex. Detects 14 deprecated primitives (RSA, ECDSA, 3DES) with zero LLM hallucination.

🧠 02. Investigate & Decide

LLM Chain-of-Thought maps vulnerabilities across the **CIA Triad**. Flags **HNDL** & calculates **PQC MTU Packet Bloat** for NIST FIPS 203/204 targets.

🔒 03. Act Safely (HITL)

LangGraph `interrupt()` halts execution. Human reviewer holds final veto. Every decision is locked in a **SHA-256 chained audit trail**.

96.8%	< 45 ms	FIPS 203/204	100%	SHA-256
AST PRECISION	STATIC SCAN LATENCY	NIST PQC STANDARDS	HUMAN-IN-THE-LOOP	CHAINED AUDIT LOG